MIDTERM EXAM

ID Number: 22104086
Group: 3

Question 1:

Given the signal $x(t)$ in Fig.1, open the file "xt.c" found in the folder named "MIDTERM" and define its piecewise representation using C language syntax by editing the function

```
double x(double t)
{
  //edit this function
}
```

To check for syntax errors in "xt.c" go back to the MIDTERM.exe prompt and follow the instructions for checking "xt.c".

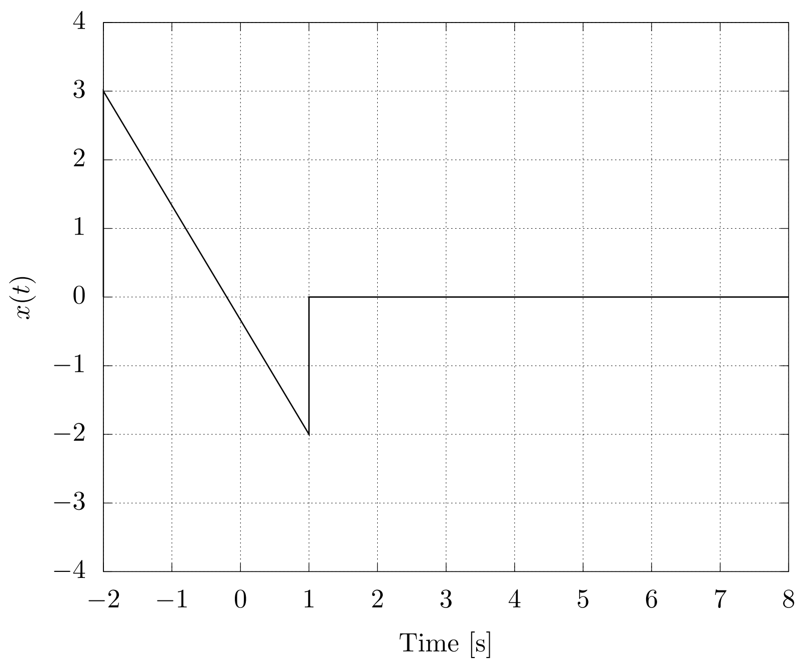

Figure 1: Signal $x(t)$

Question 2:

Given the same signal $x(t)$ from Fig.1, find the fourier transform $X(j\omega)$ of $x(t)$. Write your solution in a separate sheet of paper. Open the C file "xjw.c" and write $X(j\omega)$ as a C function with the function prototype

```c
_Complex Xj(double w);
```

where **_Complex** is data type that can accept complex numbers through the assignment operator

```c
_Complex a;
_Complex r;
_Complex i;

a = 1 + 1*j;
r = 1; //real only
i = 1*j; //imaginary only
```

The function **Xj** calculates $X(j\omega)$ based on the passed value ω. To aid in writing the correct syntax assume the following:

- The imaginary constant j is defined simply as **j**
- Imaginary expressions e.g. $-j2\pi\omega$ can be written as **-1*j*2*M_PI**
- There is a function called cexp(x) that calculates e^x where x is an expression that evaluates to a complex value i.e. containing real and imaginary parts

As an example, if one were to code $X(j\omega) = \frac{1}{-j\omega}e^{-j2\pi\omega t}$, one would define the function as

```c
_Complex Xj(double w)
{
  return 1.0/(-1*j*w)*cexp(-1*j*2*M_PI*w*t);
}
```

Use the text editor of your choice.

Question 3:
Given a discrete signal $x[n]$) shown in Fig.2,

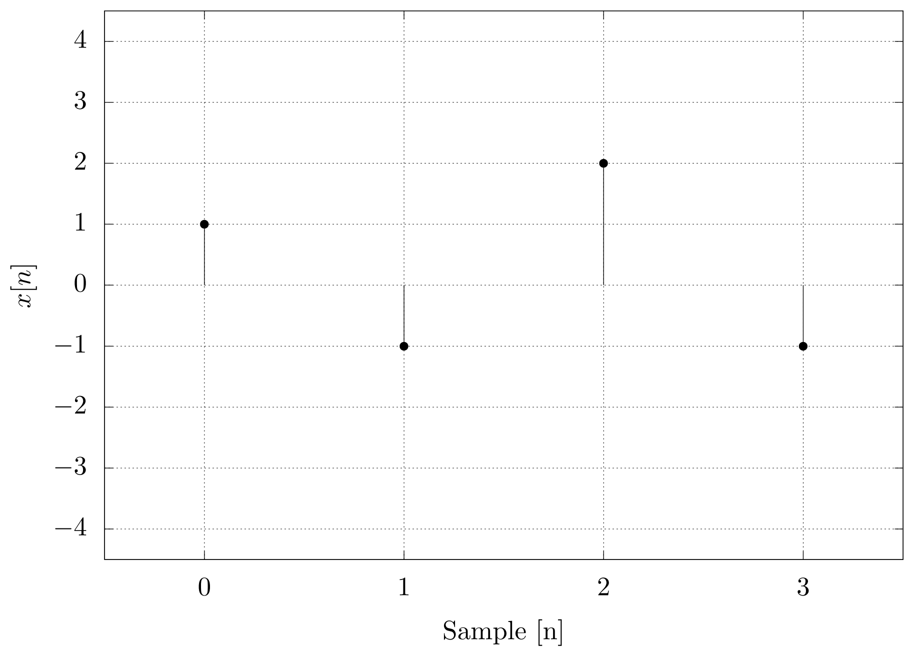

Figure 2: Signal $x[n]$

find the discrete fourier transform using

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n}. \tag{1}$$

Write your solution in a separate sheet. For the final answer, open the file named "dft.c" write the C expression for for $X[0], \ldots, X[N-1]$. Use the same assumptions as in Question 2 for the available keywords like j,π, e etcetera. As a guide, given that the expression for $X[k]$ is given to be

$$\begin{aligned}
X[0] &= 1 + e^{-j/10} + e^{-j\pi/20} \\
X[1] &= 2 + e^{-j/100} + e^{-j/200} \\
X[2] &= 3 + e^{-j/1000} + e^{-j/2000} \\
X[3] &= 0,
\end{aligned}$$

the equivalent "dft.c" file would become

```
_Complex X[4];

void dft()
{
  X[0] = 1.0 + cexp(-1*j/10) + cexp(-1*j*M_PI/20);
  X[1] = 2.0 + cexp(-1*j/100) + cexp(-1*j/200);
  X[2] = 3.0 + cexp(-1*j/1000) + cexp(-1*j/2000);
  X[3] = 0.0;
}
```

Use the text editor of your choice.

NOTE: To test all C files for possible syntax errors go back to the MIDTERM.exe prompt and follow the instructions.